

Proactive safety hazard identification using visual–text semantic similarity for construction safety management

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ABSTRACT

Automated safety management in construction can reduce injuries by identifying hazardous postures, actions, and missing personal protective equipment (PPE). However, existing computer vision (CV) methods have limitations in connecting recognition results to text-based safety rules. To address this issue, this paper presents a multi-modal framework that bridges the gap between construction image monitoring and safety knowledge. The framework includes an image processing module that utilizes CV and dense image captioning techniques, and a text processing module that employs natural language processing for semantic similarity evaluation. Experiments showed a mean average precision of 49.6% in dense captioning and an F1 score of 74.3% in hazard identification. While the proposed framework demonstrates a promising multi-modal approach towards automated safety hazard identification and reasoning, improvements in dataset size and model performance are still needed to enhance its effectiveness in real-world applications.

1. Introduction

Construction sites are inherently dangerous, with numerous safety risks that pose significant threats to workers. In the United States, the construction industry reported >1000 deaths and 75,400 nonfatal injuries in 2020 [1]. These incidents, often due to inadequate safety management and hazard identification, highlight the urgent need to address challenges such as the reliance on traditional manual observation methods. These methods are labor-intensive, prone to human error, and lack the capability for continuous monitoring.

Our study tackles the crucial issue of enhancing hazard identification processes at construction sites. Traditional approaches frequently fall short due to their manual nature and the absence of continuous monitoring. We aim to address these shortcomings by integrating computer vision (CV) and natural language processing (NLP) tools for more effective site monitoring and safety rule analysis.

To overcome these challenges, we propose a construction safety management framework that leverages CV and NLP technologies. This combination offers unique advantages: CV assists in real-time detection of unsafe conditions, such as hazardous postures [2] and the absence of

personal protective equipment [3], and NLP provides insights into safety regulations and guidelines [4]. Together, they facilitate comprehensive monitoring of behavior and interactions, as well as the identification of potential safety hazards.

Our approach fills the gap in existing systems, which typically focus on either visual cues or textual data. By merging CV and NLP, we enhance the contextual understanding gleaned from visual data and automate the interpretation of safety rule texts—key for effective safety analysis in the dynamic environment of construction sites. The proposed framework includes two modules: (1) an image processing module using CV and dense image captioning technologies to recognize behaviors and interactions, and (2) a text processing module employing NLP to extract and analyze semantic meanings from safety regulations and guidelines. This synergistic approach promises a more comprehensive and accurate hazard identification process, potentially reducing the occurrence of fatal and nonfatal injuries in the construction industry.

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2. Related works

2.1. Vision-based construction safety management

Workers' unsafe behaviors significantly result to hazards [5] on construction sites. Traditionally, Behavior-Based Safety (BBS) programs have been implemented to enhance safety performance in construction projects and organizations [6,7]. BBS is a strategy that fosters a safety partnership between management and employees by identifying and observing unsafe actions, providing direct feedback to those who engage in unsafe behaviors, and enhancing future safety awareness and performance through coaching and training [8,9]. Among these steps, the initial identification of unsafe actions is critical as it triggers the subsequent interventions. However, in practice, visually observing unsafe behavior can be labor-intensive, costly, and sometimes subjective. Moreover, documenting any potentially hazardous action or event requires filling out a Safety Observation Report (SOR), adding to the managerial and administrative burden [10].

Recently, deep learning (DL) techniques have demonstrated remarkable capabilities in visual recognition [11,12], rivaling or even surpassing human performance. These techniques are particularly adept at automatically identifying hazardous activities [9] on construction sites, including image classification, object detection, and image captioning. A visual representation (Fig. 1) of these tasks, ranked by label density and semantic richness, illustrates their effectiveness. Label density refers to the number of labels or annotations associated with an image or image segment, where a higher density indicates a more detailed identification of features within the visual data. Semantic richness, on the other hand, pertains to the depth of meaning extracted from an image, including detailed context and nuances beyond what is merely visible, ensuring a comprehensive understanding of the scene.

This shift towards automated construction safety management using CV techniques not only streamlines the process of recognizing hazardous situations but also reduces the reliance on manual observations and the subjectivity associated with them. By integrating these advanced technologies, construction sites can potentially achieve a higher level of

safety efficiency, significantly reducing the risk of accidents and enhancing overall project safety performance.

Image Classification involves the process of identifying objects or actions within an image. This technique is widely used to recognize workers, personal protective equipment (PPE), and specific construction activities. For instance, Zou and Kim [13] applied a classification algorithm to categorize excavator activities for idle time analysis. Similarly, Golparvar-Fard et al. [14] developed a support vector machine classifier algorithm for classifying activities involving earth-moving equipment. In image classification, an entire image is analyzed to produce a single label, resulting in a low label density. Typically, the output label is a single keyword, indicating that the semantic richness of image classification is also low.

Object Detection goes a step further by not only recognizing the presence of objects within an image but also identifying their categories (e.g., worker, PPE, material) and specific locations. This technique is more advanced than image classification because it processes subregions of an image. Object detection is crucial for construction safety management, enabling the identification of workers without helmets [3]; the presence of small tools at indoor sites [15]; and the safe zones around workers, man lifts, and cranes in aerial images [16]. Object detection achieves a higher label density compared to image classification but still offers low semantic richness due to its reliance on single keyword labels.

Image Captioning generates textual descriptions for images, moving beyond mere keywords to sentences that detail objects, activities, and interactions. Lu et al. [17] utilized image captioning to generate text descriptions for worker activities and interactions, and Wang et al. [18] utilized image captioning for heavy equipment working scenarios. This method provides greater semantic information about the scene, offering insights into worker activities, interactions, and equipment usage scenarios. Image captioning surpasses image classification in semantic richness, although its label density is limited by its focus on whole-image analysis.

A sophisticated vision-based safety management system necessitates CV technologies that deliver both high semantic richness and label density. Semantic richness is essential for understanding the dynamics

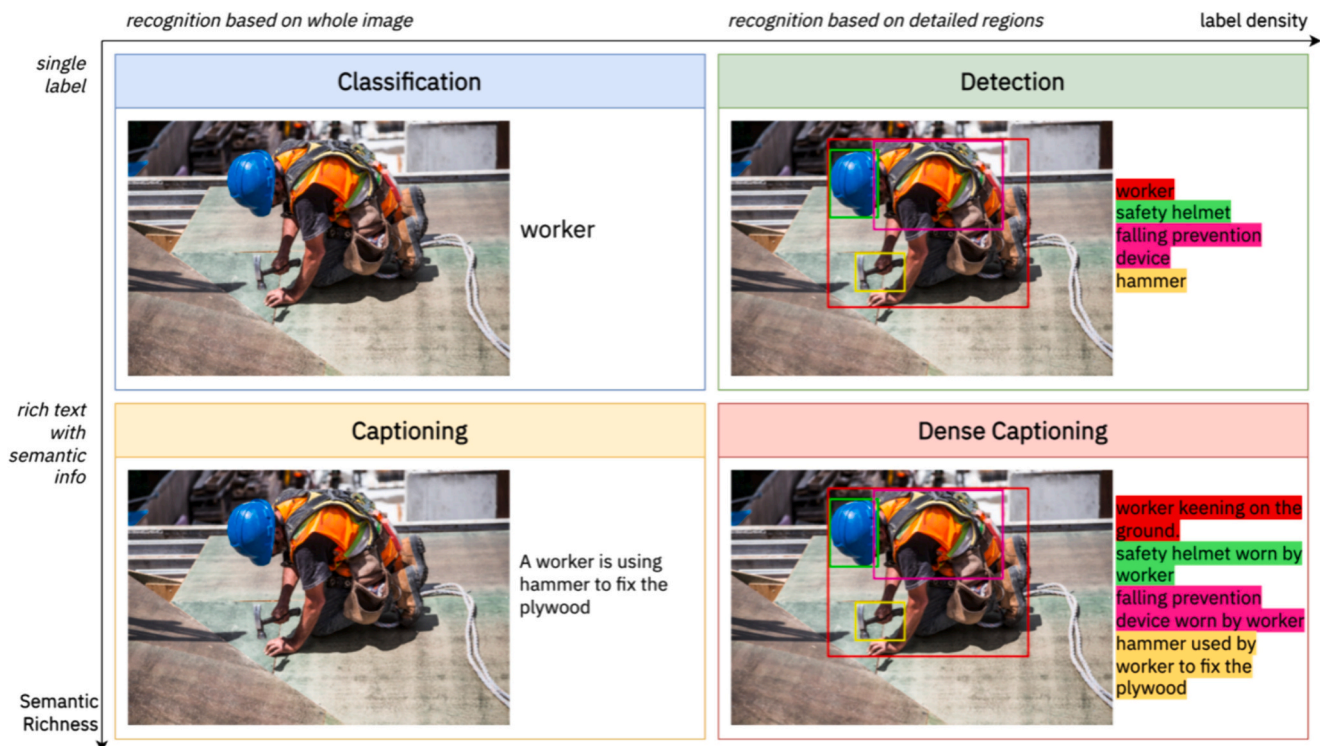


Fig. 1. Comparison of the label density and semantic richness of various visual recognition tasks.

within a scene, including how individuals and objects interact, which is critical for identifying safety risks associated with inappropriate actions and complex spatial-temporal relationships [19]. High label density is crucial for accuracy, ensuring that important scene details are not overlooked. Recent studies have aimed to produce dense and semantically rich labels. For example, Zhang et al. [20] incorporated interaction labels into an object detection model to create a scene graph. However, these labels were keyword-based rather than natural language descriptions. Dense captioning [21] emerges as a promising solution, offering the potential to generate dense and rich annotations within a single model, though it has yet to be widely adopted in vision-based safety management programs.

2.2. Natural language processing in construction safety

Data pertinent to construction safety, including safety regulations, guidelines, construction plans, safety reports, incident logs, and worker communications, is often stored in various electronic text formats. These records play a critical role in tracking construction project progress, identifying safety hazards, and fostering better communication between workers and managers. Natural Language Processing (NLP) can analyze vast quantities of text-based data, such as safety reports and incident logs, to uncover patterns and potential hazards [22]. This capability enables construction firms to identify and mitigate safety risks more effectively and efficiently, thereby enhancing safety on construction sites [23]. Similar to CV, NLP offers several techniques for processing text data, including document classification, knowledge extraction, and factor analysis, all of which can significantly improve construction safety management.

Document Classification involves sorting documents into predefined categories or labels, which can help organize construction documents like safety reports, incident logs, and progress reports. This categorization aids in pattern recognition and decision-making. For example, Tixier et al. [24] used network analysis and graph clustering algorithms to identify potential safety conflicts from injury reports. Zhang [25] proposed a deep neural network with word embeddings for automatically classifying construction accident causes. Fang et al. [26] developed a Deep Learning (DL)-based text classification method using BERT to automate the classification of near-miss incidents in safety reports.

Knowledge Extraction aims to automatically draw structured information from unstructured or semi-structured texts, extracting entities, relationships, and events, and recognizing concepts, themes, and sentiments. In construction safety management, knowledge extraction can restructure complex safety accident narratives into more understandable formats [27], extract content from safety documents to verify compliance with safety standards early on and automate the extraction of relationships describing safety requirements [28].

Text Mining uses statistical methods to find correlations and patterns in data, helping to identify underlying factors contributing to safety incidents or patterns in communications that could signal potential hazards. For instance, Kim and Kim [29] analyzed news items to identify factors leading to construction fire accidents, while Xu et al. [30] extracted safety risk factors from metro construction accident reports using text mining.

While NLP applications in construction safety have traditionally relied on text data, the growing use of digital cameras in the industry means that images and videos are becoming increasingly important for site monitoring. Combining CV and NLP offers a powerful tool for managers, providing a comprehensive view of site safety conditions, enhancing hazard identification, and improving overall safety.

For instance, Zhong et al. [31] combining image and text data by manually tagging construction scenes in images with predefined categories and using NLP to extract hazard ontologies from documents. However, they did not apply CV techniques for automatic image processing. Zhang et al. [20] took a more advanced approach by combining CV and NLP to detect safety hazards, first identifying keywords from

images and then assessing hazard probabilities using a supervised BERT model and safety regulations. Despite promising results, this approach faces limitations due to its reliance on keyword pairs with limited semantic richness and the need for extensive data collection and model training for supervised learning models.

2.3. Research gaps and objectives

The studies highlighted thus far demonstrate a significant focus on utilizing either CV or NLP techniques in the realm of construction safety management. While CV techniques are instrumental in monitoring construction sites, NLP techniques excel in analyzing text-based safety regulations and case reports. In our current research, we aim to develop an integrated method that extracts semantic information from site images and automatically identifies safety hazards.

In this research, two gaps were identified:

- 1) Current image captioning methods struggle to efficiently extract detailed information (with high label density and semantic richness) from construction images. Although some methods can label numerous objects or keywords, they fail to capture the full contextual information, lacking semantic richness. Conversely, methods capable of generating captions with richer semantics do not comprehensively label all pertinent objects and interactions, hindering a thorough analysis of construction images for potential safety issues.
- 2) Most existing methods treat image data and text data as separate entities, not leveraging the potential of integrating visual information from construction images with textual data from safety regulations. This isolated analysis of data sources diminishes the ability to identify safety hazards, which often requires cross-referencing visual contexts with applicable safety rules.

To address these gaps, the objectives of this research are as follows:

- 1) To develop an advanced dense image captioning method tailored for construction images that can produce multiple descriptive captions, achieving both high label density and semantic richness.
- 2) To develop a prototype framework that integrates construction image data and safety regulation texts. By comparing the semantic information gleaned from images against the textual safety rules, our goal is to enable the automatic identification of potential safety hazards depicted in construction images.

By achieving these objectives, we anticipate not only filling the identified research gaps but also significantly enhancing the capability of construction safety management systems to preemptively identify and mitigate safety risks through an innovative integration of CV and NLP technologies.

3. Methodology

To extract semantic information about objects, activities, and interactions, and to link this information to a domain knowledge base for identifying hazard causes, we integrated dense image captioning with NLP into a cohesive framework. A graphical depiction of the modules and submodules within our proposed framework is provided in Fig. 2. This framework comprises two main modules:

3.1. Image-to-text transformation module

This module converts image data into textual data. It starts by identifying several target regions within the construction site image, which typically include critical safety management elements like workers, personal protective equipment (PPE), and tools. Following this, the module captures the semantic information within each region,

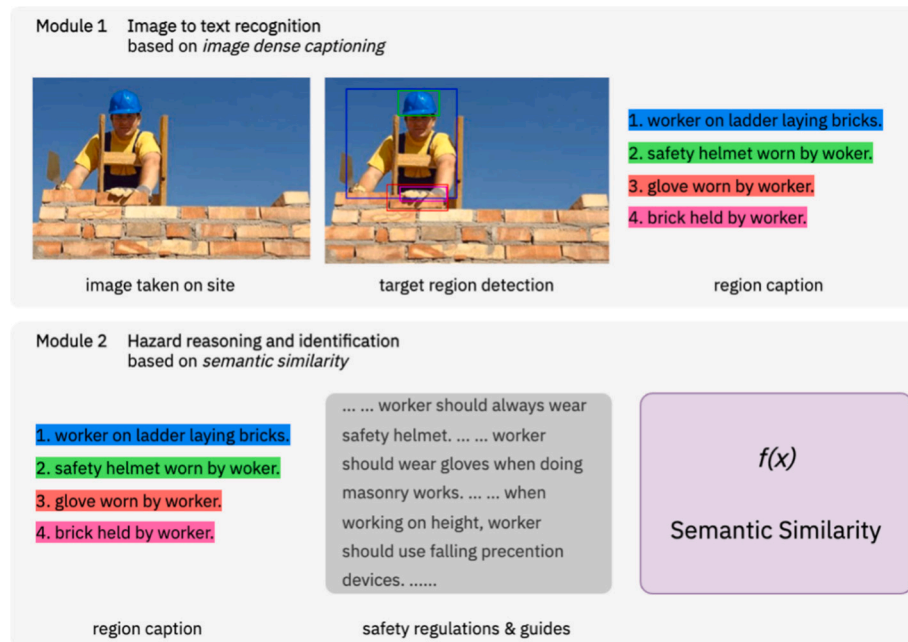


Fig. 2. Main modules of the proposed method.

articulating it in natural language captions. This ensures that essential safety-related aspects and their interactions within the image are accurately documented.

3.2. Safety hazard reasoning and identification module

The next module is dedicated to reasoning about and identifying potential safety hazards. It achieves this by evaluating the semantic similarity between the captions generated from the first module and the domain knowledge base. This module organizes the captions according to individual workers, then employs semantic similarity assessments to ascertain whether the depicted actions comply with safety regulations.

The architecture of our model is designed to facilitate easy adaptation to various construction environments, ensuring versatility. Moreover, the structure of the model is optimized to remain efficient, even as

it scales to process larger datasets, maintaining its effectiveness across different scenarios.

The computational flow of the proposed framework is illustrated in Fig. 3. Fig. 3a and b correspond to the first module and second module, respectively. Subsequent sections will delve into the specifics of each component and their computational processes, providing clarity on how our framework systematically approaches the extraction of semantic information and hazard identification.

3.3. Visual recognition and description generation

In this subsection, we describe the calculation process shown in Fig. 3a. Initially, an image of a construction site is captured and used as input. This image undergoes processing, during which semantic information is extracted in the form of text descriptors for various subregions

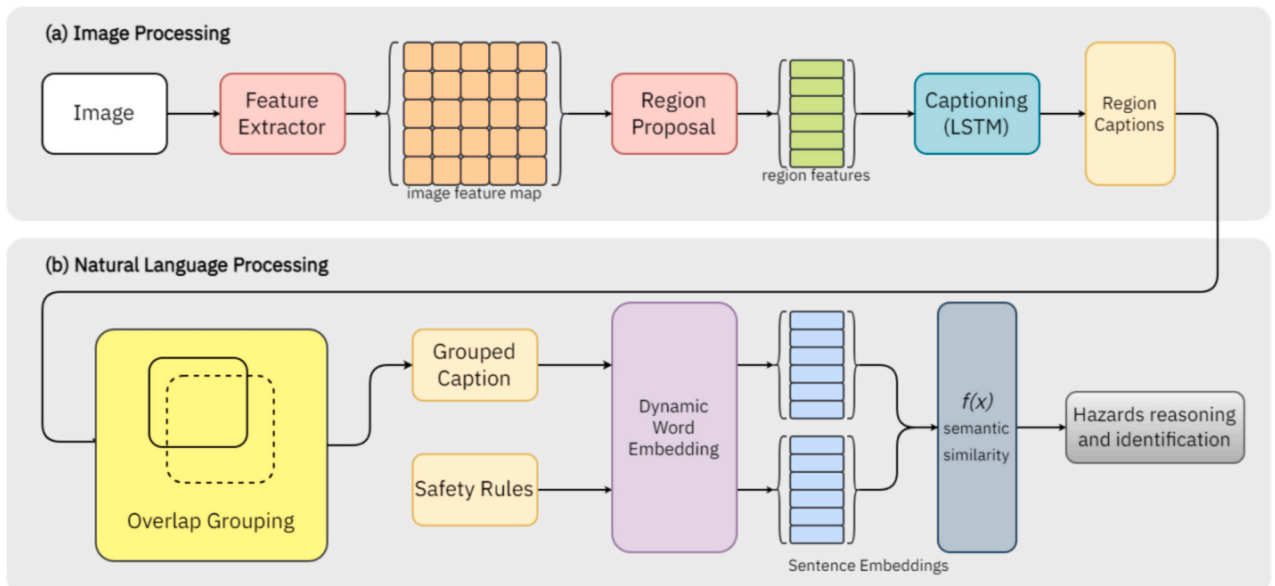


Fig. 3. Computational workflow of the proposed method.

within the image. Typically, the image data is represented by a three-dimensional (3D) tensor:

$$I \in \mathbb{R}^{3 \times W \times H}. \quad (1)$$

3.3.1. Feature extraction

The image data are first forwarded to a convolutional neural network (CNN)-based feature extraction network to obtain the image feature map. CNNs are a type of neural network designed to process grid-like input and have been used to process image data [11]. The CNN breaks down the grid data into smaller feature maps to reflect the visual properties of the image. The deeper the layer in the CNN, the higher the representation level of the image feature in the layer's output. For instance, shallow layers of the CNN identify lines and edges in the image when processing images, whereas deeper layers identify shapes, such as the overall shape of a safety helmet.

$$\mathbf{V} = \text{CNN}(I), \mathbf{V} \in \mathbb{R}^{C \times H \times W}. \quad (2)$$

where $\mathbf{V}$ is the visual feature map, C is the number of channels of the feature map, H and W are the height and width of the feature map.

3.3.2. Region proposal

The extracted image feature map is forwarded to a region proposal network (RPN) to generate target regions in the image. RPN is a CNN proposed by Ren et al. [32] and has been used for object detection. RPN uses the sliding window approach, wherein the network generates multiple subregions of the input image with different locations, sizes, and aspect ratios. These subregions have their counterparts called anchor boxes on the extracted feature map. The RPN then uses these anchor boxes to predict whether each anchor box contains an object of interest and, if so, adjusts the size and position of the bounding box to fit the object more accurately. This is accomplished using a set of regression coefficients learned during training:

$$\mathbf{B}, \mathbf{s} = \text{RPN}(\mathbf{V}), \mathbf{B} \in \mathbb{R}^{n \times 4}, \mathbf{s} \in \mathbb{R}^n. \quad (3)$$

After obtaining the subregions' coordinates $\mathbf{B}$ and scores $\mathbf{s}$, the final target subregions are filtered out by applying a threshold score, whereas the image feature vectors of the subregions are obtained using another small CNN:

$$\mathbf{v}_i = \text{CNN}(\mathbf{V}, \mathbf{B}_i), \mathbf{v}_i \in \mathbb{R}^d. \quad (4)$$

3.3.3. Dense captioning

The region features are forwarded to a long short-term memory (LSTM)-based recurrent neural network (RNN) to realize semantic information recognition and extraction. Given the constraints of our dataset size and typical captioning output of 5–10 tokens and based on empirical evidence from our prior comparative study [33], we found LSTM to be a more efficient choice in this study. Fig. 4 illustrates how

the semantic information is recognized and how the region caption is generated in the LSTM cells. The LSTM cell performs several calculations to encode the necessary information as the hidden state. The hidden state transfers the information between cells, meaning that the previous cell's hidden state is used to calculate the hidden state of the current time step.

The previous hidden state and previous output word are inputted into the LSTM cell at the current time step. Caption words are represented by one-hot vectors. The initial hidden state is the region feature vector, and the initial input word is a special word token “<start>” indicating the start of the captioning process:

$$\begin{aligned} h_t, c_t &= \text{LSTM}(h_{t-1}, x_t) \\ x_t &= c_{t-1} \text{ for } t > 0 \\ x_0 &= \text{<start>} \\ h_0 &= \mathbf{v} \end{aligned} \quad (5)$$

where t is the timestep in the LSTM decoding process, h_t is the hidden state of the decoder at time step t , x_t is the input of the LSTM at time step t , c_t is the vector representation of the output word at time step t , and $\mathbf{v}$ is the feature vector of an image subregion.

As this method is a dense captioning application, all the region features are forwarded to the LSTM cells to obtain the caption text for each region.

3.4. Caption grouping

In the process of grouping, captions are sorted into distinct clusters based on the worker they describe. Given that a single image might feature multiple workers, captions can become disorganized and not properly grouped according to the specific workers they refer to. To address this challenge, a grouping method is employed to allocate captions related to the same worker into a unified cluster. This organization ensures that all captions describing a particular worker are consolidated, facilitating easier analysis and identification of their safety status. In this study, the method of overlap measurement was selected for the grouping task.

The procedure begins with the extraction of main captions that describe the worker. During the caption annotation process, the data annotator is instructed to denote the main object in the target sub-region as the sentence subject in the caption. This implies that sub-regions focusing on workers will feature captions with “worker” as the sentence subject. Since the model's predictions are influenced by the style of the annotated captions, the output for worker sub-regions will similarly have “worker” as the primary subject of the caption. The main caption extraction process involves filtering out captions that contain the subject word “worker,” a task readily accomplished by modern NLP tools such as Spacy [34].

Following the extraction, the bounding boxes associated with the

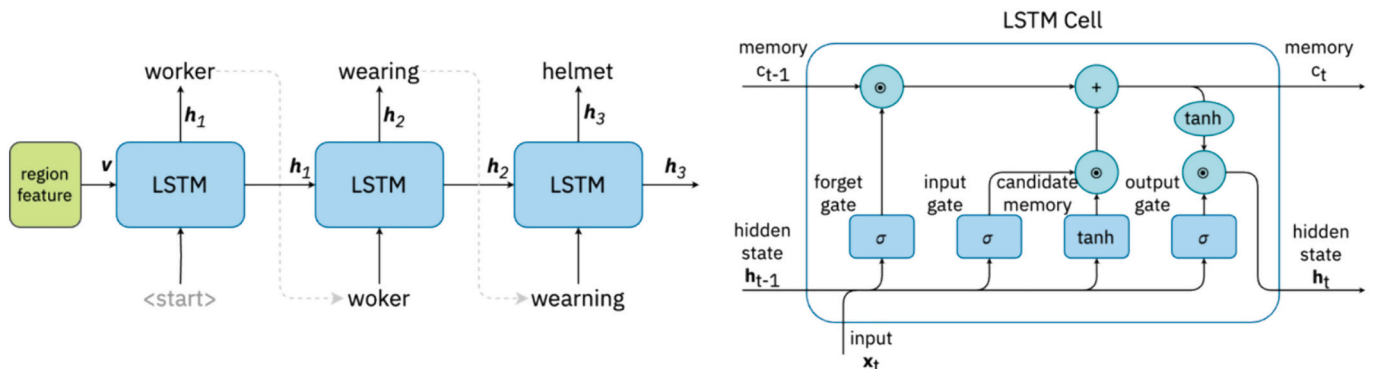


Fig. 4. Region caption generation process.

remaining captions are compared against the main bounding boxes to compute their intersection area:

$$\text{ratio} = \frac{\text{area}(\text{box}_1 \cap \text{box}_2)}{\min(\text{area}(\text{box}_1), \text{area}(\text{box}_2))} \quad (6)$$

An overlap ratio threshold λ is used to determine which main caption group the current caption belongs to. In this study, the ratio threshold was set as 0.5. Accordingly, when a caption box had an overlap ratio larger than 0.5 with multiple main boxes, the caption was assigned to the main group that has the highest overlap ratio. In contrast, when a caption box had no overlap ratio larger than 0.5 with any main boxes, the caption was regarded as redundant.

3.5. Word embedding and sentence embedding

As illustrated in Fig. 3b, the NLP-based hazard reasoning and identification process includes steps called “static word embedding” and “contextualized word embedding.” The word embedding steps are essential for the second module as they enable understanding the semantic meaning.

3.5.1. One-hot word vectorization

Traditionally, machine learning (ML) algorithms utilize one-hot vectorization to represent words. One-hot vectorization is a method used for encoding categorical data as numerical data that can be used as input to ML models, such as words in a lexicon. This method creates a new dimension for each word in the vocabulary and assigns a binary value (1 or 0 to respectively indicate the presence or absence of the word. This method produces a high-dimensional and sparse representation of the words, with most values being zero. The one-hot vectorization of several words and the two-dimensional and 3D visualization of some of the word vectors are illustrated in Fig. 5a. This method is simple and efficient for encoding categorical data but cannot represent the semantic relationship between words. As shown in Fig. 5a, all the word vectors are orthogonal to each other, providing no clue regarding the relationship between words such as synonyms and antonyms.

3.5.2. Word embedding

In contrast to one-hot vectorization, where word vectors are orthogonal, the word embedding method represents words as continuous-valued vectors in a low-dimensional space where words that have similar meanings are spatially close together. The vector representation for words is typically trained using large datasets and can capture word context as well as word relationships. A common word embedding training method is Word2Vec [35], which produces a vector space with a dimensionality of several hundred; each unique word in the corpus is allocated a matching vector in the space, as illustrated in Fig. 5b. Thus, each dimension in the vector space can be used to measure a semantic or syntactic property of the word, resulting in semantically similar words having close representation vectors.

3.5.3. Static word embedding

Word embedding based on Word2Vec is a type of static word embedding. Word vectors do not change when used with a different language corpus, and the same word is always represented by the same vector no matter which context it appears in, hence the name “static.” Word vectors are typically pretrained on a large corpus of text and then used as a fixed input for downstream tasks. The use of this type of word embedding is illustrated in Fig. 6a. The word embedding lookup table is obtained through the training process. Each time a word vector is required, the program indexes the lookup table to query the corresponding word vector. In other words, embeddings do not adapt to the specificity of sentences or datasets for which the model is being called. Thus, static word embedding represents the semantic characteristics of a word in a large context scenario and may not provide an accurate representation of the semantic relationship in the current context:

$$S = \text{Word2Vec}(C), C = [c_1, c_2, \dots, c_n]^T, \quad (7)$$

where c_i is the one-hot vectorization of the i -th word in a sentence, and $S = [s_1, s_2, \dots, s_n]^T$ is the static word embedding matrix containing static word vectors in it.

3.5.3.1. Contextualized word embedding. Contextualized word

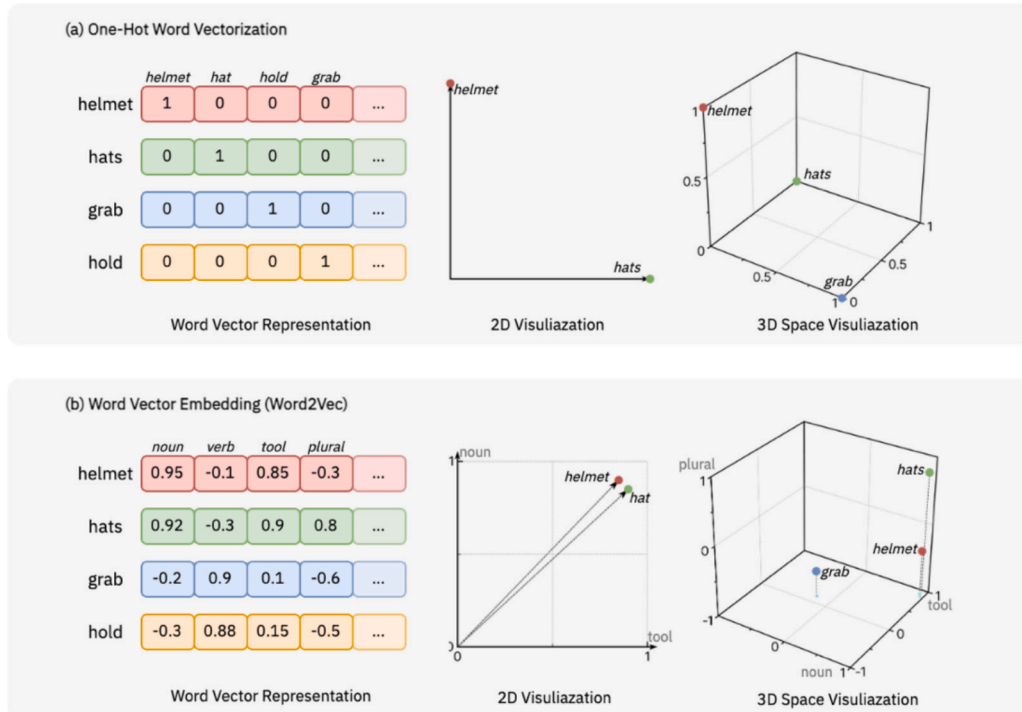


Fig. 5. Comparison between one-hot word vectorization and word embedding.

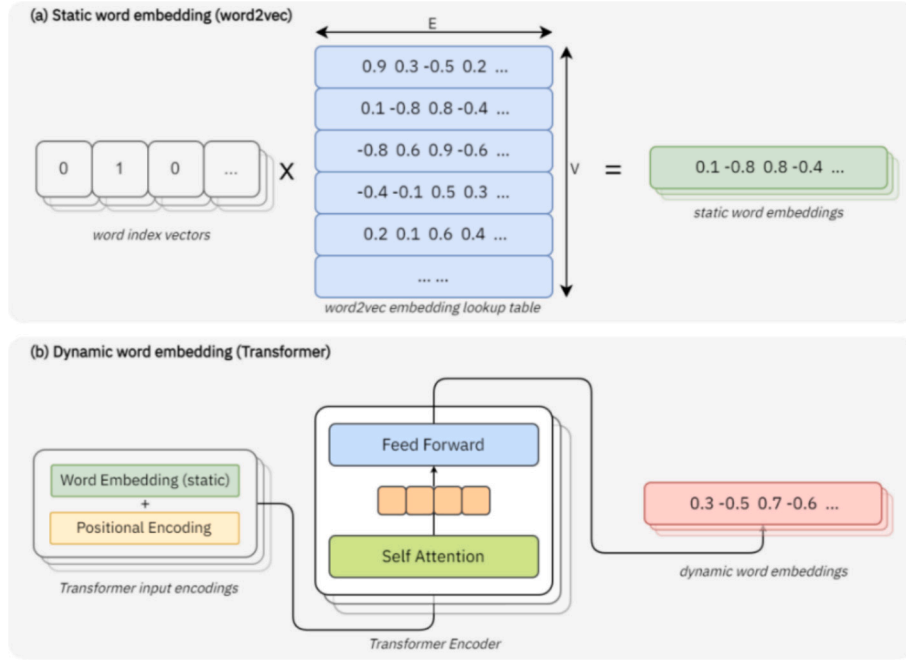


Fig. 6. Comparison between the static word embedding method based on Word2Vec and the contextualized word embedding method based on the transformer network.

embedding provides contextualized word vectors that adapt to specific sentences [36]. Contextualized embedding uses more advanced models to create representations that change depending on the context. Several contextualized word embedding models are available, such as BERT [37], universal sentence encoder (USE) [38], and Sentence-BERT [39]. These models are based on the Transformer architecture, which is a type of neural network proposed by Vaswani et al. [40]. As illustrated in Fig. 6b, contextualized word embedding is achieved using a Transformer-based language encoder. Generally, the input of the Transformer-based encoder is the static word embedding together with the positional encoding that gives the position of a word in a sentence. The Transformer-based encoder is trained on a large language corpus to learn the parameters for encoding word embeddings. During encoding, the self-attention mechanism allows the model to focus on certain parts of the input while processing it, thus enabling the model to understand the context in which words appear and generate better context-sensitive embeddings for a given sentence input:

$$\mathbf{D} = \text{Transformer}(\mathbf{S}), \mathbf{S} = [s_1, s_2, \dots, s_n]^T, \quad (8)$$

where s_i is the static word embedding of the i -th word in a sentence, and $\mathbf{D} = [d_1, d_2, \dots, d_n]^T$ is the static word embedding matrix containing static word vectors in it.

After word embeddings for the words in a sentence are generated, the vector representation of a given sentence can be obtained by applying average pooling on the word vectors:

$$\mathbf{s} = \bar{\mathbf{S}}, \mathbf{d} = \bar{\mathbf{D}}. \quad (9)$$

3.6. Universal sentence encoder

In this study, the USE [38] was used as the contextualized sentence embedding tool because the USE exhibited better performance on the data and implementation in this study. However, the USE does not outperform other methods such as BERT and Sentence-BERT in all scenarios; future research is required to explore different embedding methods.

The USE is a pretrained DL model that can generate contextualized numerical embeddings of sentences. The USE is based on a deep neural

network trained on various NLP tasks, such as sentiment analysis, paraphrase identification, and natural language inference. One of the key features of the USE is its ability to encode the semantic meaning of a sentence rather than just its surface-level syntax. This means that sentences with similar meanings are mapped to similar vectors even if they use different words or structures.

The USE is based on a deep neural network architecture called the Transformer, which was proposed by Vaswani et al. [40] for machine translation tasks and has since become a popular choice for various NLP tasks due to its ability to model long-range dependencies and capture global context.

In the USE architecture, the main component is the encoder, which is a multilayer Transformer network that processes the input sentence and generates a sequence of hidden states, one for each token in the sentence. The pooling layer then aggregates the hidden states into a fixed-length vector, which serves as the sentence embedding. The Transformer-based encoder is illustrated in Fig. 7.

The USE can generate text embeddings that can be used for various natural language understanding tasks. To achieve this, the model is trained using a multitask learning framework, where it learns to perform several tasks simultaneously. This approach helps the model learn a more general understanding of the input text, resulting in embeddings that can be used for multiple tasks. The multitask training process is illustrated in Fig. 8.

In the question-answering task (Fig. 8a), the model is trained to predict the correct answer to a given question according to the context provided. This task helps the model learn to understand the semantic relationship between questions and answers; this can be useful for other tasks that involve understanding the meaning of the text.

The Stanford Natural Language Inference dataset [41] comprises sentence pairs, where each pair is labeled as either entailment, contradiction, or neutral (Fig. 8b). The model is trained to predict the correct label for each sentence pair; this helps the model learn to recognize semantic relationships between sentences. This task enhances the model's ability to understand the meaning of the text at a sentence level; this can be useful for tasks that involve comparing sentences.

Furthermore, the model is trained on large-scale unsupervised data from sources such as Wikipedia and news articles (Fig. 8c). This training

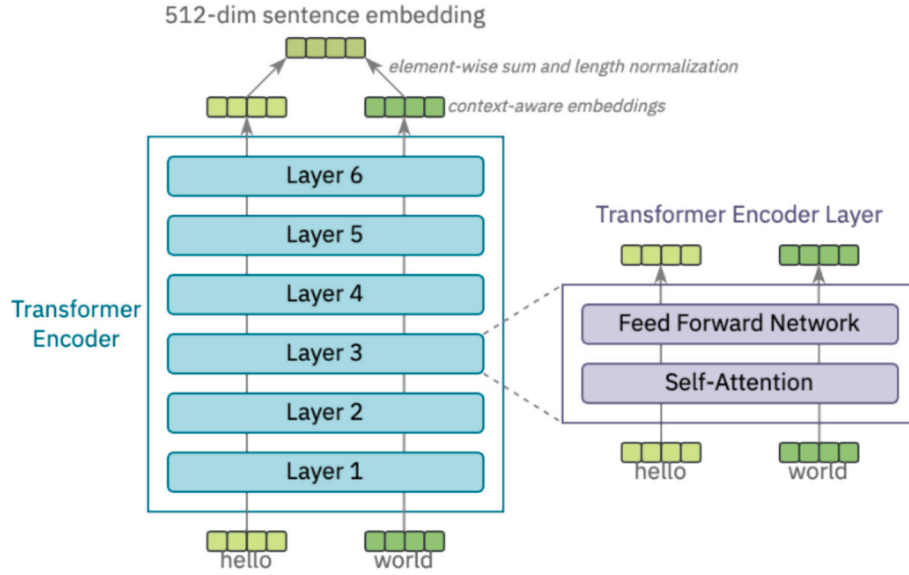


Fig. 7. Architecture of the transformer-based encoder.

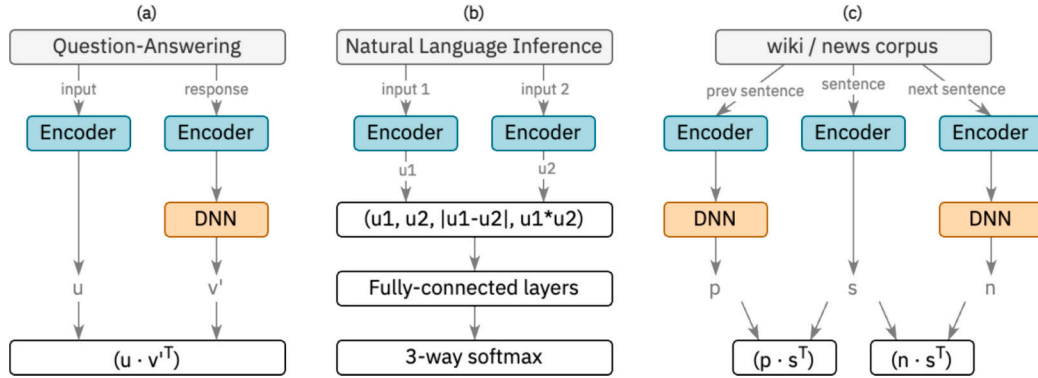


Fig. 8. Multitask training in the universal sentence encoder [38]; the tasks and task structures share the same encoder layers and parameters.

helps the model learn general language understanding and construct a broader knowledge base. Pretraining on such data exposes the model to diverse topics, writing styles, and vocabulary, thereby enabling it to generate embeddings that can be used for a wide range of tasks.

The combination of the aforementioned tasks helps the USE learn a rich understanding of text semantics, thereby allowing it to generate embeddings that capture the meaning of the input text. The final pre-training of the USE benefits from this multitask learning approach, resulting in embeddings that can be used for tasks such as text

classification, sentiment analysis, and semantic textual similarity. This is also the reason why a pretrained USE model was utilized as the contextualized sentence embedding model in this study.

3.7. Rule compliance checking

Rule compliance checking is performed by measuring the semantic similarity of the embedding vectors of captions and rules. As discussed in Section 3.3, word embedding is a technique that represents words in a

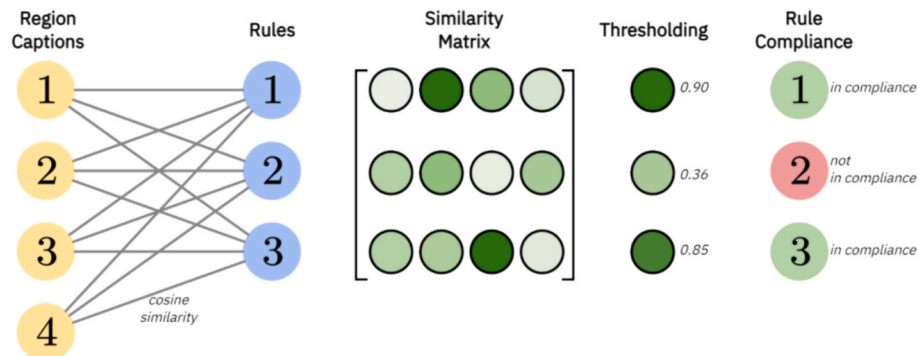


Fig. 9. Graphical illustration of rule compliance checking based on semantic similarity match.

continuous, dense vector space, where each dimension represents some semantic or syntactic feature of the word. These vectors, which in the context of this study describe the region captions and the safety rule corpus, constitute a vector space that enables the evaluation of the semantic similarity between these two textual pieces of information. This process is illustrated in Fig. 9 intuitively.

Semantic similarity is a measure of how closely related two words, phrases, or sentences are in meaning. The most common approach used for calculating semantic similarity is to calculate the cosine similarity of the word embeddings of the words or phrases in query and target texts. Given the contextualized embedding vectors of both image captions set $\mathcal{C} = (\mathbf{d}_1^{\mathcal{C}}, \mathbf{d}_2^{\mathcal{C}}, \dots, \mathbf{d}_m^{\mathcal{C}})$ and rule sets $\mathcal{R} = (\mathbf{d}_1^{\mathcal{R}}, \mathbf{d}_2^{\mathcal{R}}, \dots, \mathbf{d}_n^{\mathcal{R}})$, a similarity matrix:

$$\mathbf{S}_{m \times n} = \begin{bmatrix} s_{11} & s_{12} & \dots & s_{1n} \\ s_{21} & s_{22} & \dots & s_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ s_{m1} & s_{m2} & \dots & s_{mn} \end{bmatrix} = [s_{ij}] \quad (10)$$

can be obtained by calculating the cosine similarity:

$$s_{ij} = \text{cosin_sim}(\mathbf{d}_i^{\mathcal{C}}, \mathbf{d}_j^{\mathcal{R}})_{i=1, \dots, m; j=1, \dots, n} = \frac{\mathbf{d}_i^{\mathcal{C}} \cdot \mathbf{d}_j^{\mathcal{R}}}{\|\mathbf{d}_i^{\mathcal{C}}\| \|\mathbf{d}_j^{\mathcal{R}}\|}. \quad (11)$$

Rule compliance checking can be regarded as evaluating the semantic similarity between image captions $\mathcal{C}$ and related safety rules $\mathcal{R}$. The workflow of checking the compliance between region captions and related safety rules is displayed in Fig. 9. First, the semantic similarity s between each of the region captions and safety rules is calculated and a similarity matrix $\mathbf{S}_{m \times n}$ is obtained:

$$\mathbf{S} = \text{cosin_sim}(\mathbf{d}^{\mathcal{C}}, \mathbf{d}^{\mathcal{R}}) \quad (12)$$

Next, the semantic similarity vector $\mathbf{r}$ is calculated by applying maximum pooling on the similarity matrix:

$$\mathbf{r} = [r_j] \\ r_j = \max(s_{ij})_{i=1, 2, \dots, m}, s_{ij} \in \mathbf{S} \quad (13)$$

After obtaining the similarity vector, the safety rules that must be complied with can be selected according to a given threshold $\alpha \in [0, 1]$. If the similarity score is larger than α , then it can be selected as a safety rule that has been complied to:

$$\mathcal{R}' = \{\mathcal{R}_j\}, \text{ if } r_{j=1, 2, \dots, n} \geq \alpha. \quad (14)$$

4. Experiments

4.1. Data preparation and model implementation

We compiled a dataset based on 2000 site images provided by Zhang et al. [20], collected from a variety of real-world construction scenarios. Each image was annotated with dense image caption labels, following detailed guidelines and examples as shown in Fig. 10. The target image subregions were categorized into four main classes: human, PPE, tool, and construction materials. The annotations targeted image subregions, which were classified into four main categories: human, PPE (Personal Protective Equipment), tool, and construction materials, in line with the standard practice of validating image datasets from diverse realistic sources [42–46]. This approach is recognized for its effectiveness in preventing overfitting issues during the training of deep learning models.

For human subregions, labels were applied to depict construction workers and the actions they were performing, such as walking or lifting. PPE subregions received labels for commonly used safety equipment worn by workers, including hard hats, safety vests, protective gloves, safety goggles, face masks, and safety harnesses. Tool subregions were annotated with various hand tools prevalent in construction settings,

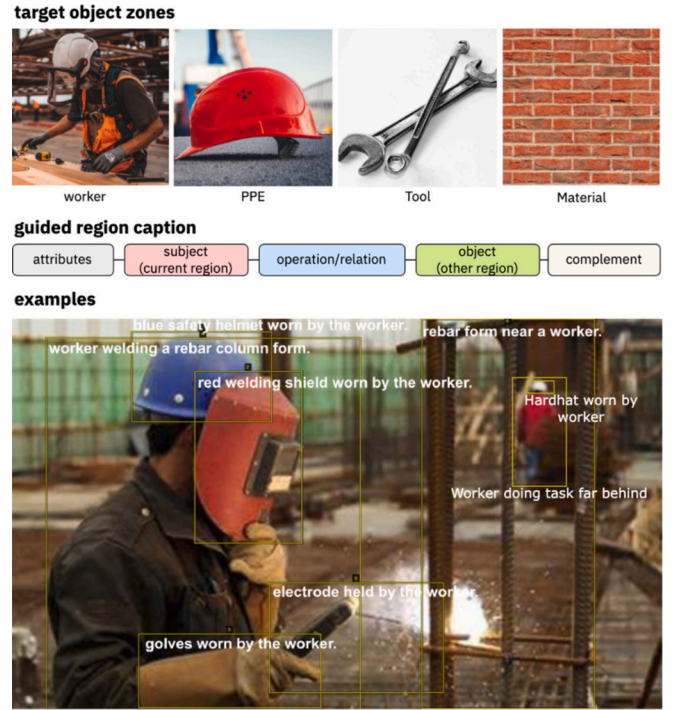


Fig. 10. Dense captioning labeling guidelines and examples.

such as hammers, wrenches, screwdrivers, power drills, and measuring tapes. Material subregions were labeled with common construction materials found on sites, like lumber, concrete, bricks, pipes, and steel beams. Objects that were not directly involved in the construction tasks, such as distant background elements, unused construction vehicles, and unhandled materials, were not included in the labeling process.

To facilitate uniformity in the captioning process, a language template was established, requiring each caption to detail the subject, operation, and object; additional attributes (e.g., color, number) and context (e.g., surrounding environment information) were also encouraged. Prior to model input, images were standardized to a resolution of 224×224 pixels and underwent RGB channel normalization, ensuring consistency across the dataset for effective model training and evaluation.

In addition, we developed an online annotation task schema to assist the two-step labeling process, as shown in Fig. 11. In the first step, the labelers were asked to draw bounding boxes for a set of target object categories. In the second step, after each bounding box was annotated on the image, a query popped up asking the labeler to provide a text description for the drawn bounding box, providing semantic information for the subregion.

As a result, over 10,000 image subregions and captions were labeled. As illustrated in the “example” section of Fig. 10, each image in the dataset has an $n \times 4$ bounding box annotation matrix indicating the subregion coordinates, where n is the number of subregions. Each image also contains a list of strings, which contains n captions indicating the captions of each of the subregions. Some statistics about the dataset and the data distribution are provided in Table 1.

We implemented the proposed method in a Python environment by using the workflow shown in Fig. 12. In addition to the prepared and annotated image dense captions, we prepared several safety rule sets as presented in Table 2. The rules are considered according to the activities included in the image dataset, and fellow studies [45,47]. After labeling the caption dataset for the construction images, the dataset was randomly split into the training set and validation set in the 9:1 ratio. The training set has 1800 images and the validation set has 200 images. The dense captioning model was then trained and fine-tuned. We

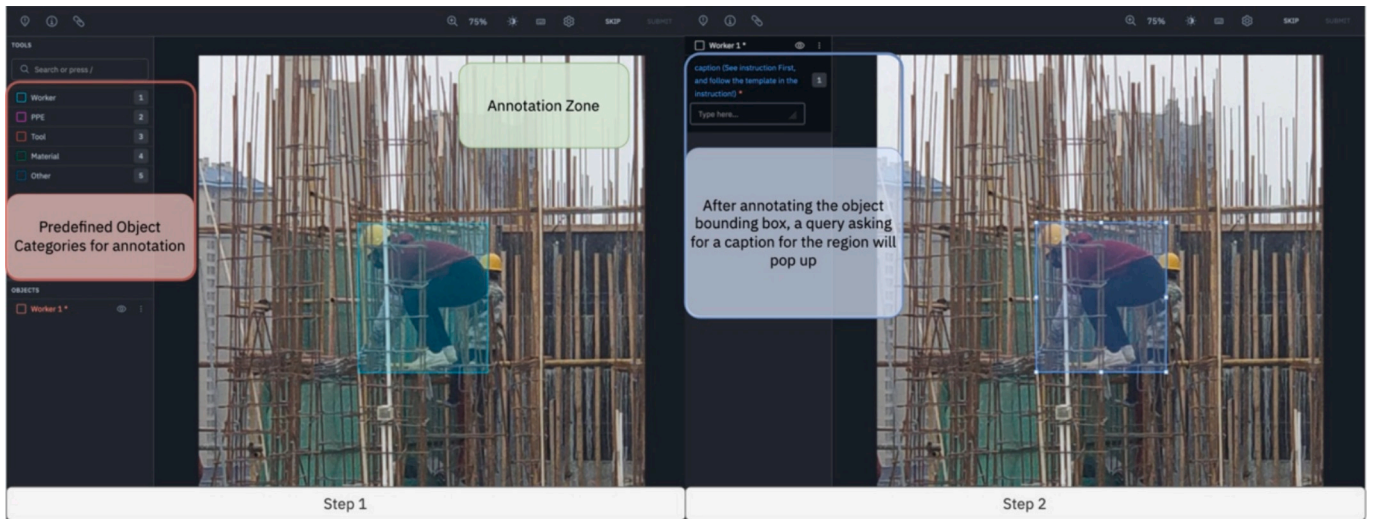


Fig. 11. Annotation platform and schema.

Table 1
Dataset statistics.

Statistic Item	Detail
Object Numbers	worker: 7266 ppe: 3267 tool: 1212 material: 790 other: 107
Total Objects/Total Captions	12,642
Average length of captions	4.59 words
Most 10 frequent words	worker: 12,364 occurrences by: 8004 occurrences worn: 6674 occurrences safety: 3153 occurrences helmet: 3018 occurrences yellow: 2032 occurrences gloves: 1421 occurrences on: 1379 occurrences glove: 1376 occurrences the: 1321 occurrences black: 5 occurrences flag: 3 occurrences forklift: 7 occurrences vests: 2 occurrences jacket: 29 occurrences camera: 2 occurrences tarpoline: 2 occurrences drone: 2 occurrences machinery: 30 occurrences hoe: 3 occurrences
Least 10 Frequent Words	bricking: 314 images concreting: 284 images height working: 714 images rebar work: 349 images other: 339 images

constructed this model based on the proposed method by using Python libraries such as PyTorch [48] and TorchVision [49]. We used a pre-trained ResNet-50 network [50] as the image feature extracting network as ResNet is a commonly utilized feature extractor. The captions and related rules sets were then fed into the NLP part for sentence embedding and semantic similarity evaluation. For the NLP, we developed the module by using Python libraries such as HuggingFace Transformer [51] and spaCy [34]. Furthermore, we used Word2Vec as the static word embedding method and the USE as the contextualized word embedding model. In addition, Table 3 provides some key hyper-parameters in the model:

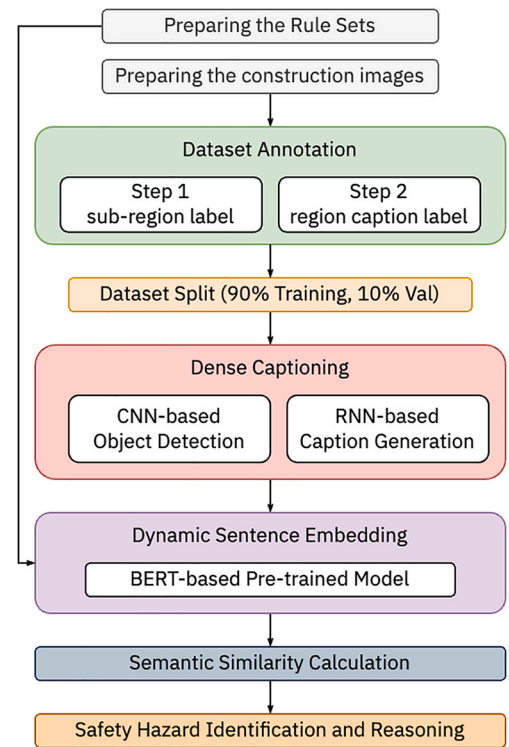


Fig. 12. Implementation workflow.

Table 2
Safety rules used in this study.

Ruleset	Activity Type	Regulations
1	Working at a height	1. Worker should wear a helmet. 2. Worker should wear fall-prevention device.
2	Concrete	1. Worker should wear foot-protection boots. 2. Worker should wear gloves. 3. Worker should wear helmet.
3	Brick-laying	1. Worker should wear a safety helmet. 2. Worker should wear gloves.

Table 3
Key hyperparameters used in this model.

Hyperparameter	Value
Caption feature size	4096
LSTM hidden size	1024
Caption maximum length	12
Caption embedding size	512
Learning rate	0.0001
Batch size	4

4.2. Evaluation metrics

Two objectives have to be achieved when generating regional captions by using the proposed module: (1) generating well-localized target regions (in object detection tasks); and (2) generating accurate descriptions (in image captioning tasks). In this study, we utilized the same evaluation metric used in the original dense captioning study [21] to text the performance of the proposed captioning module. This metric is based on mean average precision (mAP) and Metric for Evaluation of Translation with Explicit Ordering (METEOR).

mAP is a common metric used to evaluate the performance of object detection models. It is calculated as the mean of the average precision (AP) for different threshold levels. The threshold is defined as intersection of union (IoU), which quantifies the overlap between the ground-truth (GT) detection bounding box and the predicted (PD) box:

$$\text{IoU} = \frac{\text{area}(GT \cap PD)}{\text{area}(GT \cup PD)} \quad (15)$$

An IoU threshold $\alpha \in [0, 1]$ is used to distinguish the true positive (TP), false positive (FP), and false negative (FN) detections. For example, if $\alpha = 0.5$, then a detected bounding box should have an IoU larger than 0.5 with the ground truth to be considered as a correct detection (i.e., TP). Based on the IoU threshold, for each image in the validation dataset, the model's predicted bounding boxes are compared with the ground truth annotations to create a set of true positive (TP), false positive (FP), and false negative (FN) detections, and the precision and recall are calculated accordingly:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (16)$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (17)$$

AP is the mean of the precision at different recall levels:

$$AP_\alpha = \frac{1}{11} \sum_{r \in \{0.0, 0.1, \dots, 1.0\}} \text{Precision}_\alpha(r) \quad (18)$$

mAP is the average of AP at different IoU thresholds:

$$\text{mAP} = \frac{1}{5} \sum_{\alpha \in \{0.3, 0.4, 0.5, 0.6, 0.7\}} AP_\alpha \quad (19)$$

METEOR [52] is a metric used to evaluate the quality of image captioning models. It is a variant of the BLEU [53] metric, which is commonly used in machine translation. METEOR is based on n-gram overlap between the predicted and reference captions but also considers synonyms and paraphrases by using a stemming algorithm and WordNet, a large lexical database of English. METEOR is a more sophisticated and robust method than BLEU for evaluating image captioning models as it considers synonyms, paraphrases, and word order. In this study, we also used a METEOR threshold $\varepsilon \in \{0, 0.05, 0.1, 0.15, 0.2, 0.25\}$ to distinguish TP, FP, and FN. In addition, we used another mAP score mAP_{cap} to measure the AP across all pairwise settings of IoU thresholds α and METEOR thresholds ε .

When conducting checking of the result of activity recognition, hazard identification, and hazard reasoning, F1 metric was also used:

$$F1 = 2 * \text{Precision} * \text{Recall} / (\text{Precision} + \text{Recall}). \quad (20)$$

5. Results and discussion

5.1. Model predictions and evaluation results

Examples of the dense captioning output obtained using the model are shown in Fig. 13; only the target subregions with top 10 confidence are shown, and the regional captions are sorted according to the confidence rate. The evaluation results for mAP are presented in Table 4. The inference speed of the model was around 15 images/s on a single RTX 3080 Laptop GPU. The proposed model exhibited an mAP of 58.9% when evaluating only the target-region localization. An et al. [54] reported a benchmark mAP of 50.64% for a Faster R-CNN model (ResNet50) on the MOCS dataset. This means that the object detection performance of the our model is feasible. When the METEOR threshold was considered, the proposed model yielded an mAP_{cap} of 49.6%.

The accuracies of activity classification, hazard identification, and rule reasoning were evaluated. As can be seen from the results presented in Table 5, the precision scores of the activity classification were 97.0%, 95.2%, and 56.1% for bricking, height working, and spreading concrete, whereas for hazard identification, the precision scores were 76.2%, 61.4%, and 69.5%, respectively. The rule reasoning accuracy was evaluated to determine whether the model correctly predicted the safety rule that had been violated. As can be seen from the analysis results, the corresponding score was slightly lower than the hazard identification precision. In addition, some examples of safety hazard identification and rule reasoning are provided in Table 6.

In conclusion, leveraging image captioning and semantic similarity to automatically identify safety hazards in construction images shows significant promise. However, this approach is not without its limitations. Detailed discussions of failure cases and the primary reasons for errors are explored in subsequent sections, underscoring the need for further research and refinement to boost the method's accuracy and efficiency.

Despite these challenges, our evaluation results affirm the viability of combining image captioning with semantic similarity analysis to pinpoint safety hazards effectively. The system has proven capable of detecting a substantial number of hazards present in the images, underscoring its potential to improve safety management practices on construction sites. Our assessments further indicate encouraging adaptability across different scenarios, suggesting that with access to larger datasets, the model's ability to recognize activities and relationships could be significantly enhanced.

However, it is important to note that for complex safety regulations, relying exclusively on semantic similarity might not be entirely sufficient. This highlights the necessity for incorporating more nuanced analytical techniques or supplementary data sources to accurately interpret and apply intricate safety rules, ensuring a comprehensive and effective hazard identification process.

5.2. Failure cases

We identified three major causes for the errors: (1) false image captioning, which occurs when the system generates a caption that does not accurately describe the contents of the image; (2) missing image captioning, where the system fails to provide a caption for an object; and (3) over-similar word embedding, which occurs when the system cannot distinguish between words in the construction domain. Examples of failure cases along with the associated errors that caused these failures are presented in Table 7.

5.2.1. False image captioning

This error arises when the system generates captions that inaccurately describe the image content. The implications of such errors vary; when inaccuracies pertain to main worker behaviors, they can lead to incorrect safety rule applications, significantly contributing to activity



Fig. 13. Examples of the dense captioning outputs of the site images.

Table 4

Evaluation scores for the mean average precision of the dense captioning model.

Model	mAP (%)	mAP _{cap} (%)	Note
This Study	58.9	49.6	
An et al. [54]	50.6	–	This model is an object detection model trained on construction-related datasets.
Johnson et al. [21]'s model, reproduced by [55]	25.3	9.4	This model is a dense captioning model trained on general image dataset. The reproduced version is coded with the same deep learning package and evaluation scripts with this study.

Table 5

Activity recognition, hazard identification, and reasoning accuracy.

Task Type	Precision (%)	Recall (%)	F1 (%)
Overall Activity Recognition	78.9	90.4	84.3
Bricking Activity Recognition	97.0	94.1	95.5
Concreting Activity Recognition	56.1	95.8	70.8
Height Activity Recognition	95.2	80.0	87.0
Overall Hazard Identification	69.03	80.9	74.3
Bricking Hazard Identification	76.2	82.8	79.3
Concreting Hazard Identification	69.5	78.1	73.6
Height Hazard Identification	61.4	81.8	70.1
Overall Hazard Reasoning	58.8	76.0	66.0
Bricking Hazard Reasoning	76.7	85.0	80.6
Concreting Hazard Reasoning	43.0	59.5	49.9
Height Hazard Reasoning	56.6	83.3	67.4

classification errors. Conversely, false captions related to objects or other image regions can distort semantic similarity matching, potentially identifying irrelevant hazards or overlooking actual ones. This undermines both the accuracy of safety hazard identification and the reasoning behind safety rule applications.

5.2.2. Missing image captioning

Occurs when the system omits captions for crucial objects (e.g., PPE) or significant subregions. This omission is particularly problematic when it misrepresents compliance with safety protocols as hazards, due to the absence of relevant captions. Such errors are notably impactful in specific activities, like “spreading concrete,” where they substantially reduce the accuracy of hazard identification. For example, the low precision and recall for Concreting Hazard Reasoning in Table 5 mainly due to the failure to caption safety boots worn by a worker and false predictions of foot protection hazards.

The two aforementioned causes of error identified in this study, that is, false image captioning and missing image captioning, are limitations of the image captioning module. The occurrence of false image captioning and missing image captioning imply that the image captioning module may not have been trained on a sufficiently diverse set of construction site images containing numerous potential safety hazards. These limitations can be addressed by improving the efficiency of the image captioning module through the optimization of the model architecture or by increasing the size and diversity of the training dataset. Incorporating more data instances into the training dataset will improve the model's ability to accurately describe safety hazards in various contexts.

5.2.3. Over-similar word embedding

This issue arises when the system fails to differentiate between semantically similar words, particularly noticeable with words describing PPE items. Due to the functional and semantic similarities among many PPE items, the model's word embedding module may generate closely aligned word vectors for these terms, complicating semantic similarity assessments. Addressing this challenge might involve fine-tuning the similarity thresholds or, more effectively, training the word embedding model on a construction-specific corpus. This tailored approach would help the model distinguish between closely related terms within the construction context, such as different types of PPE, enhancing the accuracy of semantic similarity matching and reducing errors caused by over-similar word embeddings.

These identified errors underscore the complexity of automatically identifying safety hazards in construction images and suggest pathways

Table 6

Examples of image captioning and safety hazard identification.

Image	Captions	Unsafe?	Violated Rule
	["worker laying clay bricks on the wall," "gloves worn by the worker," "brick held by the worker"]	TRUE	["worker should wear a safety helmet"]
	["worker laying clay bricks on the wall," "yellow safety helmet worn by the worker," "gloves worn by the worker," "brick held by the worker"]	FALSE	[]
	["worker spreading the concrete mix," "yellow safety helmet worn by the worker," "rubber boots worn by the worker," "rubber boots worn by the worker," "concrete spreader held by the worker," "gloves worn by the worker," "rubber boots worn by the worker"]	FALSE	[]
	["worker spreading the concrete mix," "wheelbarrow pushed by the worker," "gloves worn by the worker"]	TRUE	["worker should wear a safety helmet," "worker should wear safety boots"]
	["worker creating a scaffold at a height," "yellow safety helmet worn by the worker," "gloves worn by the worker"]	TRUE	["worker should wear fall-prevention device"]
	["worker creating a scaffold at a height," "yellow safety helmet worn by the worker," "fall-prevention device worn by the worker"]	FALSE	[]

for refining the system's accuracy and reliability. Implementing these improvements could significantly advance the system's capability to support safety management on construction sites.

5.3. Discussion

5.3.1. Feasibility

The integration of image captioning and semantic similarity techniques offers a highly promising avenue for automated safety management in the construction sector. Advances in computer vision and

natural language processing have paved the way for image captioning models to generate accurate textual descriptions from construction scenes. Semantic similarity methods further enable the comparison of these captions with safety regulations to reliably identify rule violations. Our framework leverages the strengths of both approaches, creating a prototype capable of automated safety hazard identification and reasoning. The success of dense captioning models tailored for construction environments, combined with sophisticated context-aware similarity matching, underscores the potential of this method for proactive hazard identification, as demonstrated by our experimental results in activity recognition, hazard identification, and safety reasoning.

5.3.2. Inaccuracies

Despite confirmed feasibility, inaccuracies in image captioning systems, particularly false and missing captions, can significantly affect safety applications in construction. Such errors risk misidentifying hazards and misinterpreting safety rules, potentially leading to overlooked hazards or unnecessary interventions. Addressing these inaccuracies, perhaps through training on a larger dataset or fine-tuning for specific scenarios, is crucial for improving the systems' accuracy and reliability, thereby bolstering their contribution to safety standards maintenance.

5.3.3. Reliance on dataset

The model's efficacy heavily depends on the quality and diversity of its training dataset. A dataset that encompasses a broad spectrum of construction environments, activities, and safety scenarios is vital for the model's ability to accurately recognize and caption diverse real-world situations. Without this diversity, the model's generalization capabilities may be compromised, leading to inaccuracies in unfamiliar settings. The dataset's quality, in terms of image resolution and caption accuracy, is equally critical for developing a dependable system. A lack of quality or diversity can result in a biased or limited system, undermining its effectiveness in safety monitoring.

5.3.4. Deployment challenges

Implementing the proposed method in the dynamic, complex settings of real-world construction sites poses significant challenges. The inherent variability and constant evolution of construction environments require a highly adaptable and robust system capable of accurately interpreting various scenarios. Moreover, environmental factors such as dust, noise, and weather variations can impact image quality and caption accuracy, necessitating continuous system calibration and an up-to-date dataset reflective of construction realities.

5.3.5. Choice of architecture

The selection of the captioning architecture is influenced by the specific needs, scenarios, and constraints of the application. Future research might explore alternative decoding architectures, such as Transformer, to better meet these specific requirements.

5.3.6. Novelty of integration

Our framework presents unique benefits over methods that rely solely on computer vision or NLP. While CV techniques lack the ability for semantic reasoning, and NLP methods cannot process visual data, our integrated approach harnesses both for enhanced activity understanding and data processing. The use of unsupervised similarity matching adds flexibility for adapting to changing safety regulations without the need for retraining. Nevertheless, challenges remain in accurately perceiving highly complex scenes and sparse language, highlighting the potential of multi-modal approaches and areas for further improvement.

5.4. Methodological and practical contributions

The proposed method integrates visual and textual semantic

Table 7
Examples of failure cases.

Cause	Image	Captions	Unsafe?	Violated Rule
False image captioning (False caption indicated by underlining)		[“ <u>fall-prevention device worn by the worker</u> ,” “worker creating a scaffold at a height,” “yellow safety helmet worn by the worker”]	False	N/A
Missing captioning (Missing caption for rubber boots)		[“gloves worn by the worker,” “worker spreading the concrete mix,” “yellow safety helmet worn by the worker,” “gloves worn by the worker,” “ <u>concrete mix vibrator held by the worker</u> ”]	TRUE	[“worker should wear safety boots”]

information for the identification of safety hazards, offering several methodological and practical advantages:

5.4.1. Methodological advantages

5.4.1.1. Enhanced semantic information through dense captioning. The method utilizes dense captioning to extract detailed regional descriptions from site images, offering a significant improvement over traditional visual recognition models. It provides outputs with higher label density and semantic richness, delivering comprehensive semantic information about objects, actions, and interactions, including their localization, crucial for effective safety management.

5.4.1.2. Integration of visual and text data. By transforming image data into text-based regional captions containing semantic information and utilizing word embeddings to process these captions, the method effectively bridges the gap between visual and text data. This integration is pivotal in developing vision-based safety management programs that can seamlessly utilize both types of data.

5.4.1.3. Decoupling of visual recognition and safety hazard identification. Unlike existing methods, this approach allows for the separate optimization of the visual and language modules without altering the overall architecture. The language module, being unsupervised, maintains its robustness amidst changes in safety rules or the introduction of new regulations, offering flexibility and adaptability in safety hazard identification.

5.4.2. Practical contributions

5.4.2.1. Automation of construction safety management. Traditional safety management, reliant on manual observations, is both costly and inefficient. By employing computer vision technologies for automated site monitoring, this method enhances efficiency and reduces costs associated with construction safety management.

5.4.2.2. Enhancement of vision-based safety programs. The method's ability to automate hazard identification through the integration of semantic information extraction techniques enables more efficient safety-related decision-making. This improvement allows managers to address complex safety hazards more effectively, enhancing the overall safety management process.

5.4.2.3. Applicability to broader construction management tasks. Beyond

safety hazard identification, the dense captioning and visual-text semantic similarity techniques have potential applications in various construction management areas. For instance, regional descriptions from dense captioning could automate the generation of daily site reports or serve as metadata for image archives, facilitating advanced information retrieval from image databases. Additionally, the visual-text semantic similarity technique could verify the consistency between manual observation reports and image monitoring records, ensuring accuracy and reliability in construction documentation.

These contributions underscore the method's potential to revolutionize safety management practices in the construction industry, demonstrating its utility not only in enhancing safety but also in broadening the scope of automated construction management applications.

5.5. Limitations and recommendations

The proposed method, while innovative, has certain limitations and areas for improvement that need to be addressed:

5.5.1. Dataset diversity

The current dataset, though carefully curated, covers only a limited set of construction activities and scenarios. It lacks representation of unique construction practices, environments, or conditions found in different regions, which may limit the model's ability to generalize across all construction scenarios globally. Expanding the dataset to include a more diverse range of construction methodologies and environments from various regions could enhance the model's robustness and applicability.

5.5.1.1. Incorporation of construction safety knowledge. The dense image captioning process relies on generic labels and does not specifically consider construction safety knowledge, such as insights from incident reports. Incorporating domain-specific information, such as natural language processing of regulations and incident reports, could improve the accuracy and richness of the generated captions, thereby enhancing hazard recognition capabilities. Future developments should focus on integrating advanced multi-modality technologies to embed construction safety knowledge more deeply.

5.5.1.2. Safety rulesets and NLP techniques. The study utilized simplified safety rulesets, which may not fully capture the complexity of original safety regulations. Exploring other NLP techniques to parse complex safety regulations into usable rulesets is necessary. Adopting sentence simplification tasks (split-and-rephrase) in NLP could aid in this effort,

enabling the translation of complex regulatory language into clear, actionable safety rules.

5.5.1.3. Word embedding models. The Word2Vec-based static model and the BERT-based contextualized model used in this study were designed for general purposes and might not capture specific semantic meanings and relationships critical in the construction safety context. Training word embedding models on a construction-related corpus could significantly improve NLP performance in construction scenarios, ensuring more accurate semantic analysis and hazard identification.

5.5.1.4. Logic and reasoning abilities. The current approach is based on simple, unambiguous safety rules and lacks the ability to handle complex, context-dependent regulations. Future work could involve integrating dense image captions with large language models (LLMs) like GPT-3, known for their strong reasoning and inference capabilities. This integration could enable the models to assess compliance and identify risks with greater depth and nuance, moving beyond simple similarity matching to a more comprehensive understanding of safety regulations in context.

Addressing these limitations and exploring these areas for improvement would significantly enhance the effectiveness, accuracy, and applicability of the proposed method for automated construction safety management.

6. Conclusion

This paper introduced an innovative framework designed to automatically identify potential safety hazards in construction images by leveraging the synergy of image captioning and semantic similarity techniques. Our primary goal was to devise a methodology capable of processing visual data from construction sites to effectively pinpoint risky behaviors or unsafe conditions through comparison with textual safety knowledge bases.

The experimental outcomes underscore the efficacy of our approach in extracting detailed semantic information from construction images via dense image captioning and in precisely detecting violations of regulations by assessing the similarity between captions and safety rules. The framework exhibited classification accuracy scores ranging from 75% to 93% in recognizing hazardous activities, and it achieved over 80% accuracy in identifying and reasoning about safety hazards across most types of activities. Notably, the model demonstrated adaptability to diverse construction settings and hinted at potential scalability with the inclusion of larger datasets. Nonetheless, to fully comprehend and apply complex safety rules, further integration with sophisticated NLP methodologies is suggested.

This research underscores the practicality of merging computer vision with natural language processing to forge an automated system dedicated to proactive safety management within the construction domain. The proposed method adeptly utilizes a multi-modal understanding of both visual and textual data to pinpoint risks potentially overlooked by manual inspections, thereby facilitating early hazard identification to prevent accidents and enhance worker safety in this inherently high-risk field.

Although our study lays a significant groundwork and offers a proof-of-concept for subsequent research endeavors, it acknowledges limitations, such as inaccuracies in captioning within complex scenes and a deficiency in advanced reasoning for intricate regulations. As the construction sector progresses, continuous refinement and adaptation of our deep learning model are crucial to maintain its relevance and effectiveness in addressing the evolving landscape of construction practices and safety challenges. Future directions include expanding the image dataset, enhancing language processing capabilities, and fine-tuning the system to better manage the complexities of on-site environments. With further advancements, the combined use of image analytics and

semantic knowledge analysis promises substantial practical benefits in automating safety protocols within the construction industry.

CRediT authorship contribution statement

Yiheng Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Data curation, Conceptualization. **Bo Xiao:** Writing – review & editing, Supervision, Project administration, Data curation, Conceptualization. **Ahmed Bouferguene:** Supervision, Project administration, Methodology, Conceptualization. **Mohamed Al-Hussein:** Supervision, Resources, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data or models that support the methods and findings of this study are available from the corresponding author upon reasonable request. In addition, the code and corresponding data preparation process are available on: <https://github.com/structron/ConDenseCap>.

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